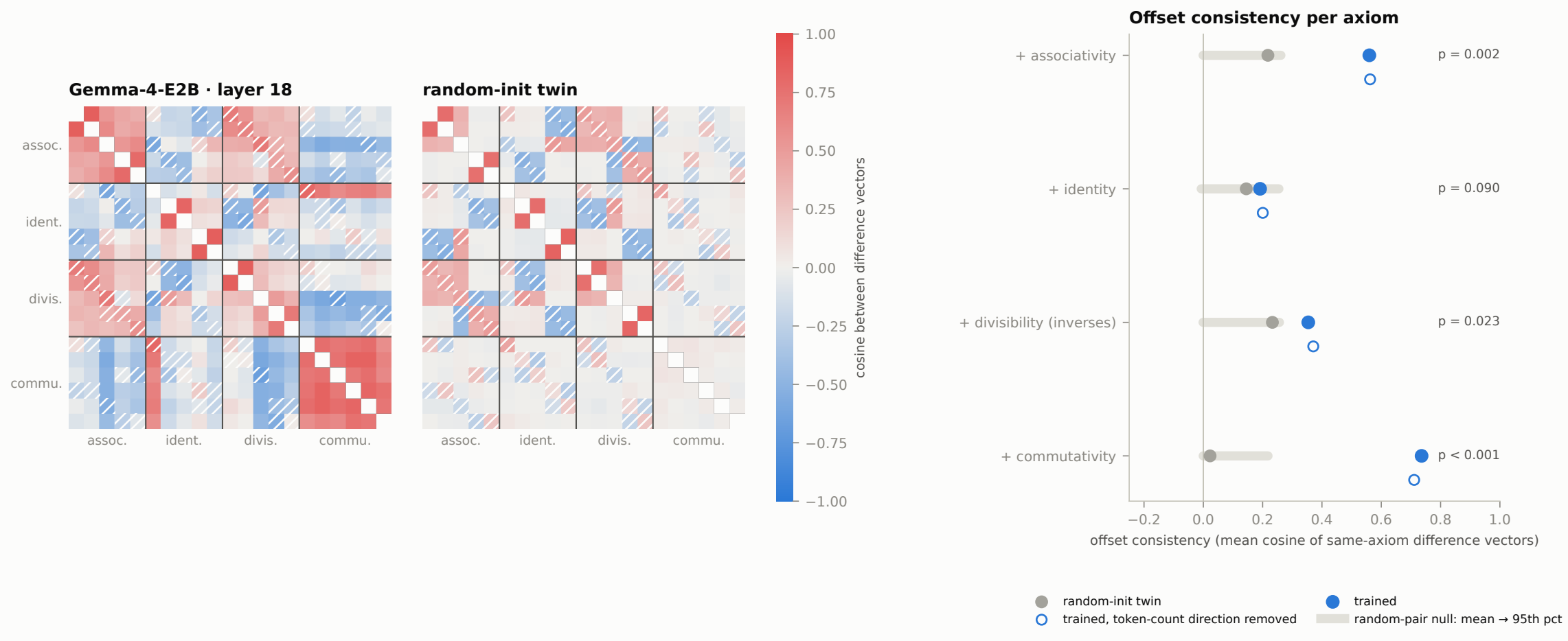
Gemma-4-E2B · layer 18
random-init twin
assoc.
ident.
divis.
commu.
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
cosine between difference vectors
Offset consistency per axiom
+ associativity
+ identity
+ divisibility (inverses)
+ commutativity
p = 0.002
p = 0.090
p = 0.023
p < 0.001
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
offset consistency (mean cosine of same-axiom difference vectors)
random-init twin
trained
trained, token-count direction removed
random-pair null: mean → 95th pct